Course Code	Course Name	Course Structure
CSCSC01	Advanced Data Structures and Algorithms	L-T-P 3-0-2

COURSE OUTCOMES(COs):

CO1: To understand the design and analysis of advanced data structures such as hash tables and priority queues.
CO2: To analyze balanced search tree structures and their operations for efficient data organization.
CO3: To design and apply graph algorithms for solving shortest path and connectivity problems.
CO4: To apply disjoint set data structures for solving equivalence and connectivity problems efficiently.
CO5: To implement and analyze string matching algorithms for pattern searching applications.

COURSE CONTENT:

Unit-I:

Hashing: General idea, Hash function, Separate chaining, Hash tables without linked lists, Linear probing, Quadratic probing, Double hashing, Rehashing, Hash table in standard library, Universal hashing, Extensible hashing.

Unit-II:

Priority Queue: Model, simple implementation, Binary heap, Structure property, Heap order property, Basic heap operations - insert, delete, percolate down and other heap operations, Binomial queues - structure and operations, Implementation of binomial queue, Priority queue in standard library.

Unit-III:

Multi Binary Search Trees: 2-3 Trees, Searching for element in 2-3 trees, Inserting new element in 2-3 trees, Deleting element in 2-3 trees, Red-Black trees - properties, rotation, insertion, deletion, AVL trees - single rotation, double rotation, B-Trees.

Unit-IV:

Graph Algorithms: Elementary graph algorithms, Topological sort, Single Source Shortest Path (SSSP), Dijkstra's algorithm, All pair shortest path, Floyd-Warshall algorithm.

Unit-V:

Disjoint Sets: Equivalence relation, Basic data structure, Simple union and find algorithm, Smart union and path compression algorithm. String Matching: Naive string matching algorithm, Robin-Karp algorithm, Knuth-Morris-Pratt (KMP) algorithm.

Text Books:

[T1] Mark Allen Weiss, “Data Structures and Algorithm Analysis”, Pearson.
[T2] Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, “Introduction to Algorithms”, MIT Press.

References:

[R1] Coursera.org/learn/advanced-datastructures.
[R2] ocw.mit.edu/6-851S12.
[R3] NPTEL Courses.
[R4] MOOC-List.com/search/node?key=Advanced+Data+Structures.